

CSA0711
SIMATS ENGINEERING
ASSESSMENT TOOL 3

39

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 30%
Software: Cisco Packet Tracer, Wireshark

Code of Conduct:

I T. Subahan (192511121) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Subahan

Criteria	Weightage	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks	Q5
Problem Identification	20%	2	1	2	1	1
Debugging	25%	1	1	1	1	1
Optimization	20%	1	1	1	1	1
Analysis	20%	1	1	1	1	1
Code Quality	15%	1	1	1	1	1

28/30

28/30

Annexure A

Debugging & Optimisation Task

1. A network administrator has configured a workstation with the following settings:

- IP Address: 192.168.10.65
- Subnet Mask: 255.255.255.192 (/26)
- Default Gateway: 192.168.10.1

However, the workstation is unable to communicate with other systems configured in the subnet 192.168.10.0/26.

Task

Debugging and Optimization of IPv4 configuration:

IP Address : 192.168.10.65

Subnet Mask : 255.255.255.192 (126)

Default Gateway : 192.168.10.1

Identify the Subnet:

Subnet mask 126 means:

Block size = 64

Subnets are:

1. 192.168.10.0/26

2. 192.168.10.64/26

3. 192.168.10.128/26

4. 192.168.10.192/26

The IP address 192.168.10.65 belongs to : 192.168.10.64/26

Network details:

Network address : 192.168.10.64

Broadcast address : 192.168.10.127

Valid host range : 192.168.10.65 to 192.168.10.126

Verify IP and gateway:

IP address:

192.168.10.65 is valid

Gateway:

Gateway = 192.168.10.1

192.168.10.1 belongs to subnet:

Device A Network :

2001 : db8 : 0 : 0 : 64

Device B Network :

2001 : db8 : 0 : 0 : 164

Same Subnet Verification

2001 : db8 : 0 : 0 : 164

Addressing mismatch :

No Subnet mismatch exists

- Firewall blocking traffic
- Incorrect gateway
- Duplicate addresses
- Interface down

Optimized configuration :

Device A :

2001 : db8 : 0 : 0 : 10/64

Device B :

2001 : db8 : 0 : 0 : 20/64

3. Subnet optimization using VLSM :

Network : 192.168.1.0/24

Subnet 1 (126) :

- Network : 192.168.1.0
- Broadcast : 192.168.1.63
- Host range : 192.168.1.1 - 192.168.1.62
- Usable hosts : 62

Subnet 2 (127) :

- Network : 192.168.1.64
- Broadcast : 192.168.1.95
- Host range : 192.168.1.65 - 192.168.1.94

corrected addresses

→ Device A : 2001:db8:0:0::10/64

→ Device B : 2001:db8:0:0::20/64

Available hosts:

$$2^{64} = 18,446,744,073,709,551,616$$

5. Debugging the Routing Table:

<u>Destination</u> - <u>Network</u>	<u>Next Hop</u>
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192.168.1.0/24	10.0.0.2
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192.168.2.0/24	10.0.0.3
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Default Route	10.0.0.1
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Packet destination:

192.168.2.100

Longest - Prefix match:

Destination :

192.168.2.100

matching route :

192.168.2.0/24

Next Hop Selected:

10.0.0.3

Verify Subnet:

→ Network : 192.168.2.0/24

→ Broadcast : 192.168.2.255